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Github : <https://github.com/bathukashivashankar/Forecasting-German-Electricity-Demand>

Electricity Demand Forecasting for Germany: A Time Series Case Study

1. Introduction

Electricity demand forecasting is critical for power system planning, market operations and reserve scheduling because forecast errors directly affect system reliability and operating costs. Accurate load forecasts allow operators to balance supply and demand, schedule generation efficiently and prepare for periods of unusually high demand.

This study evaluates several forecasting approaches for German electricity demand, starting from simple statistical benchmarks and progressing to more advanced time-series, feature-based and neural network models. The aim is not only to identify the most accurate model, but to explain **why** some methods perform better or worse on this series, recognising that complex models do not automatically outperform simpler ones when the data have strong and stable seasonality. Advanced models are therefore compared against credible benchmarks, and the results are interpreted in relation to both the characteristics of the data and the modelling choices implemented in code.

The dataset consists of German electricity load from 1 January 2015 to October 2020, obtained from the Open Power System Data time-series collection. The main analysis uses a weekly aggregation for Parts 1–5 and an hourly/daily representation for the LSTM in Part 6. A training period from early 2015 to late 2018 and a test period covering the last two years are used so that all models are evaluated on unseen data in a realistic forecasting setting.

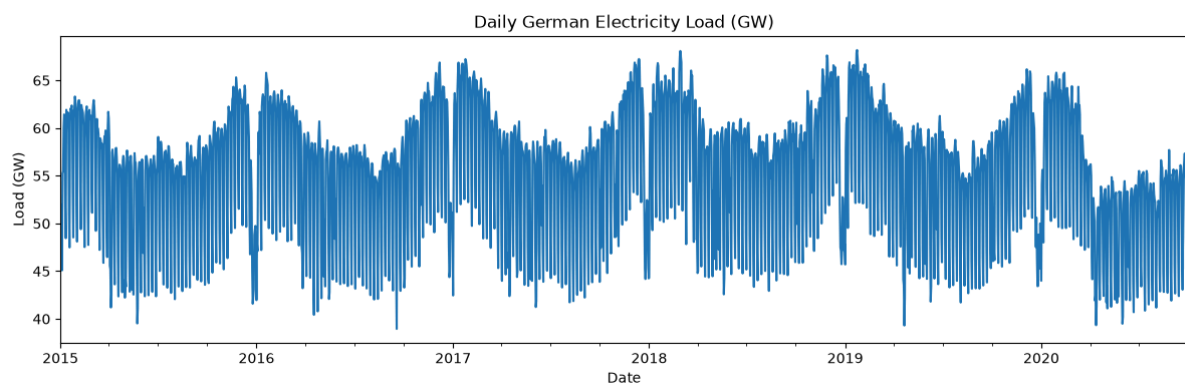


Figure 1. Weekly German electricity demand from 2015–2020 with train/test split marked (weekly_load.png).

2. Data and forecasting problem

The weekly dataset forms a univariate time series of German electricity demand measured in gigawatts, indexed by week. Visual inspection shows pronounced annual seasonality with recurring winter peaks and lower summer demand, as well as medium-term changes in level, implying that seasonal structure is a major source of predictability. Daily and hourly versions of the series exhibit similar patterns with additional short-term variability.

The forecasting task for Parts 2–5 is defined as one-step-ahead weekly prediction over a 104-week test period, using only information available up to each forecast origin and, where relevant, lagged or engineered features. This design avoids information leakage and ensures that reported metrics reflect genuine out-of-sample performance rather than in-sample fit. Accuracy is assessed using mean absolute error (MAE), root mean squared error (RMSE) and bias; MAE measures typical error magnitude, RMSE penalises large misses more strongly, and bias indicates systematic over- or under-prediction. For the LSTM model, a daily horizon over the same calendar period is considered, and aggregated metrics are compared with the weekly models.

In addition to load, Berlin temperature is retrieved from the Open-Meteo archive and aggregated to weekly averages, providing an exogenous covariate for SARIMAX and the feature-based Random Forest. Calendar variables such as week-of-year and year are also constructed, because electricity demand is strongly shaped by both weather and recurring calendar effects, which can explain departures from the basic seasonal pattern that purely univariate models may miss.

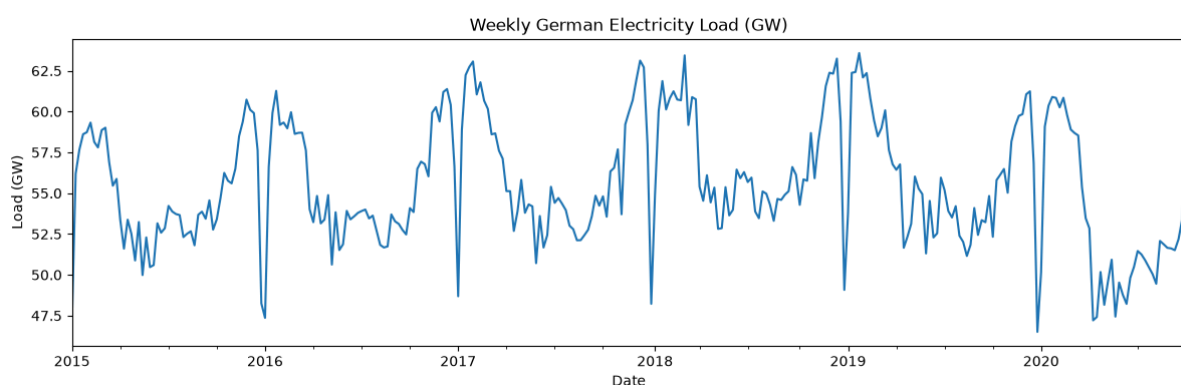


Figure 2. Daily and weekly load series illustrating strong annual seasonality and level changes (daily_load.png, weekly_load.png).

3. Methods

The modelling strategy follows a progression from simple and interpretable baselines to more flexible models. This is essential for a good forecasting study because advanced methods should only be considered useful if they beat credible benchmarks, not merely because they are more sophisticated.

3.1 Benchmark models

Four benchmark methods are used: mean, naïve, seasonal naïve and drift. The mean model predicts every future value as the historical average, so it provides a weak baseline for checking whether the series contains any exploitable temporal structure. The naïve model uses the last observed weekly value as the next forecast, while the drift model extrapolates a linear trend from the beginning to the end of the training sample.

The most important benchmark is the **seasonal naïve** model, which forecasts each week using the value from the same week in the previous year, with a seasonal period of 52 weeks. This is particularly relevant for electricity demand because the series has strong annual seasonality, and forecasting literature consistently shows that seasonal naïve is often difficult to beat when seasonality is stable. In this project, it serves as the minimum standard that the more complex models should improve on.

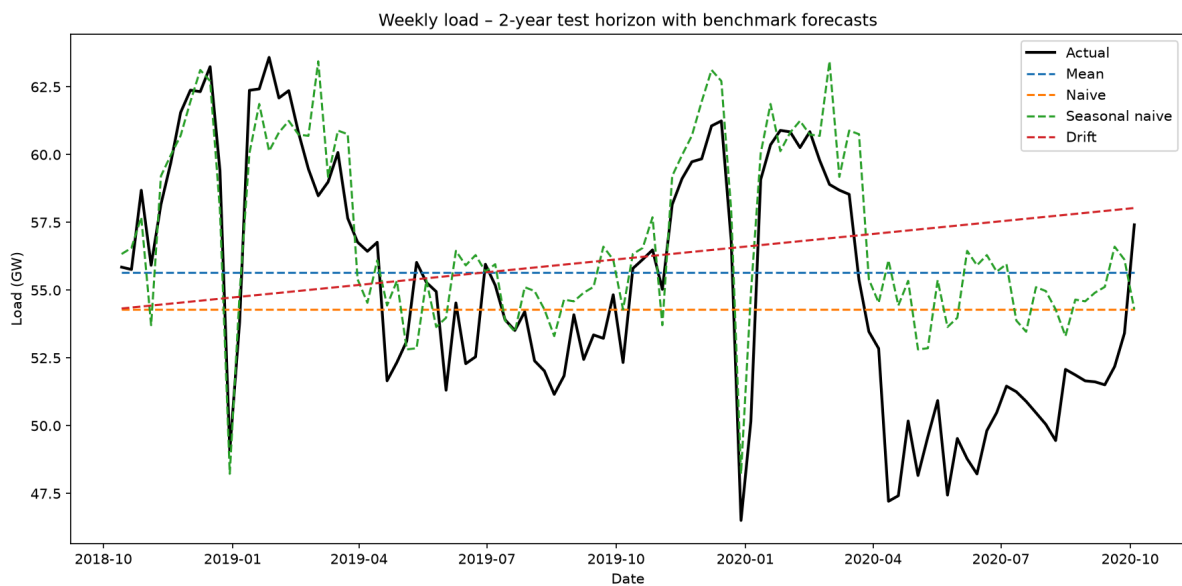


Figure 3. Seasonal naïve forecast and other benchmarks versus actual demand in the 2-year weekly test period (benchmarks_weekly_2y.png).

3.2 SARIMA and SARIMAX

A seasonal autoregressive integrated moving average (SARIMA) model is then used to explicitly model autocorrelation and annual seasonality in the weekly demand series. SARIMA is appropriate because it can represent both short-term persistence and recurring seasonal effects through autoregressive, differencing and moving-average terms. After exploratory analysis, a model with first-order non-seasonal and seasonal differencing and seasonal period 52 is selected via an AIC grid search over candidate orders.

If exogenous variables are included, the model becomes SARIMAX, allowing demand to depend not only on its own history but also on covariates such as temperature. The motivation for using SARIMA/SARIMAX is that electricity demand is not purely random; it shows repeated seasonal patterns and serial dependence, and these are exactly the patterns that the model is designed to capture. Compared with the seasonal naïve method, SARIMA can adapt better if the seasonal pattern

changes gradually over time, while SARIMAX may improve further if external variables explain unusual weeks that are not well captured by past load alone.

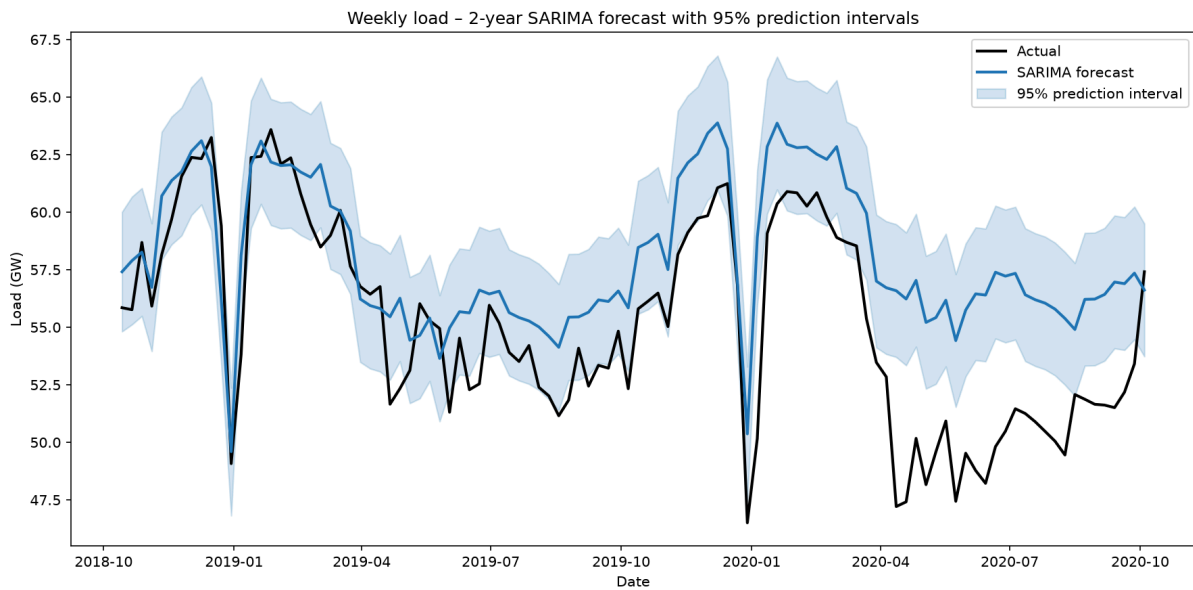


Figure 4. SARIMA forecast versus actual weekly demand with 95% prediction intervals (*sarima_weekly_2y_ci.png*).

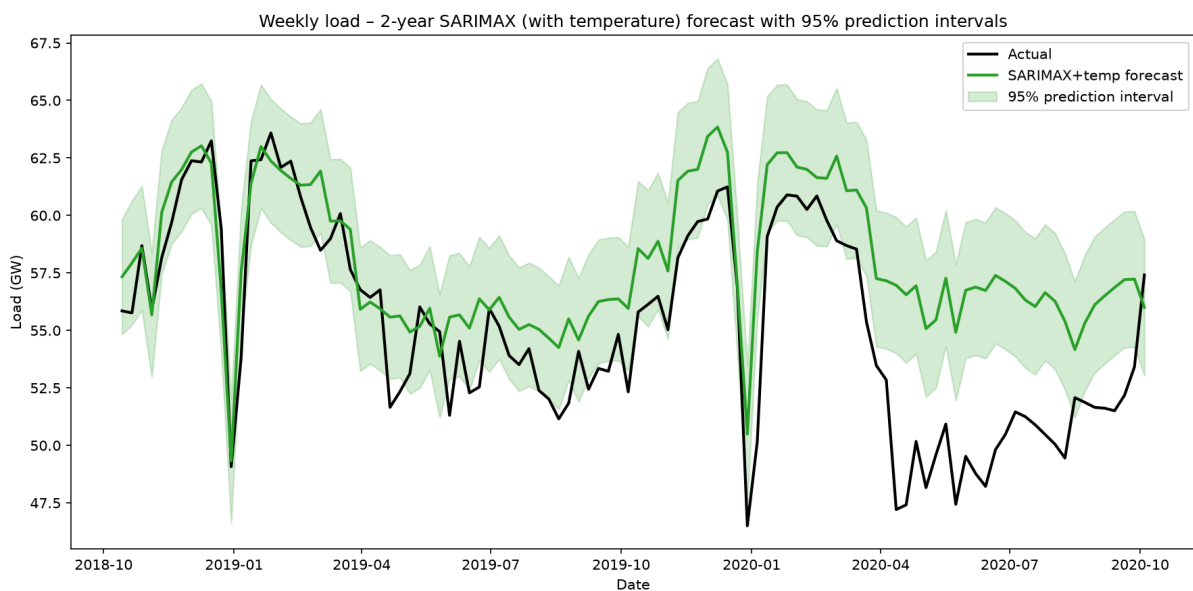


Figure 5. SARIMAX with temperature covariates versus actual demand with 95% prediction intervals (*sarimax_temp_weekly_2y_ci.png*).

3.3 Feature-based Random Forest

To test whether nonlinear methods add value, the analysis includes a feature-based Random Forest regressor using weekly data. Engineered predictors include lagged load values, lagged and contemporaneous temperatures, and calendar indicators such as week-of-year and year. Random Forests can capture nonlinear interactions and variable-specific effects without assuming linearity, which is useful when the influence of temperature or season is not constant across the year.

The model is trained on the weekly feature matrix with the last two years reserved as a holdout set. Because all lag and calendar features are computed using information available strictly up to each forecast origin, no future information leaks into the training data. The fitted model then generates one-step-ahead forecasts over the test window.

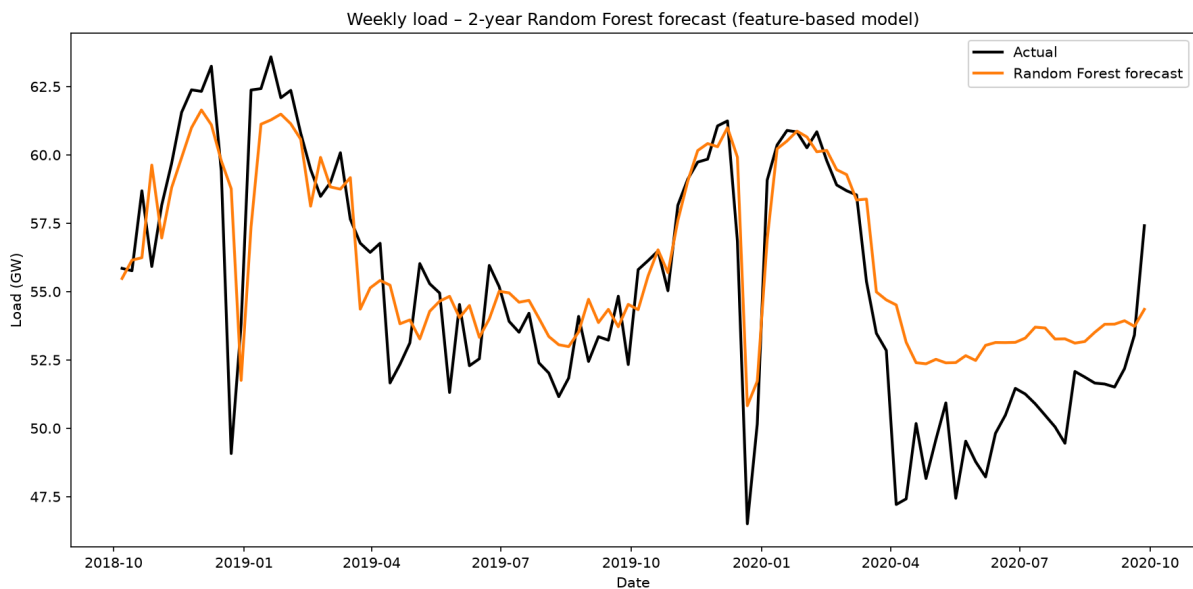


Figure 6. Random Forest weekly forecasts compared with actual demand in the test period (*rf_weekly_2y.png*).

3.4 LSTM neural network

Finally, an LSTM (long short-term memory) neural network is fitted to **hourly data aggregated to daily** demand to explore whether sequence-based deep learning can capture additional temporal structure. LSTMs are widely used in load forecasting because they are designed to learn complex, nonlinear dependencies from sequences of past observations.

A univariate LSTM architecture is implemented in Keras, using sliding windows of past daily load to predict the next day, with a two-year daily test horizon aligned with the weekly holdout period. Hyperparameters such as number of layers, hidden units, dropout and learning rate are tuned in a Google Colab environment due to TensorFlow compatibility constraints on the local machine. The final model, selected via validation loss, is then evaluated on the fixed test period, and the resulting forecasts and metrics are exported back into the project.

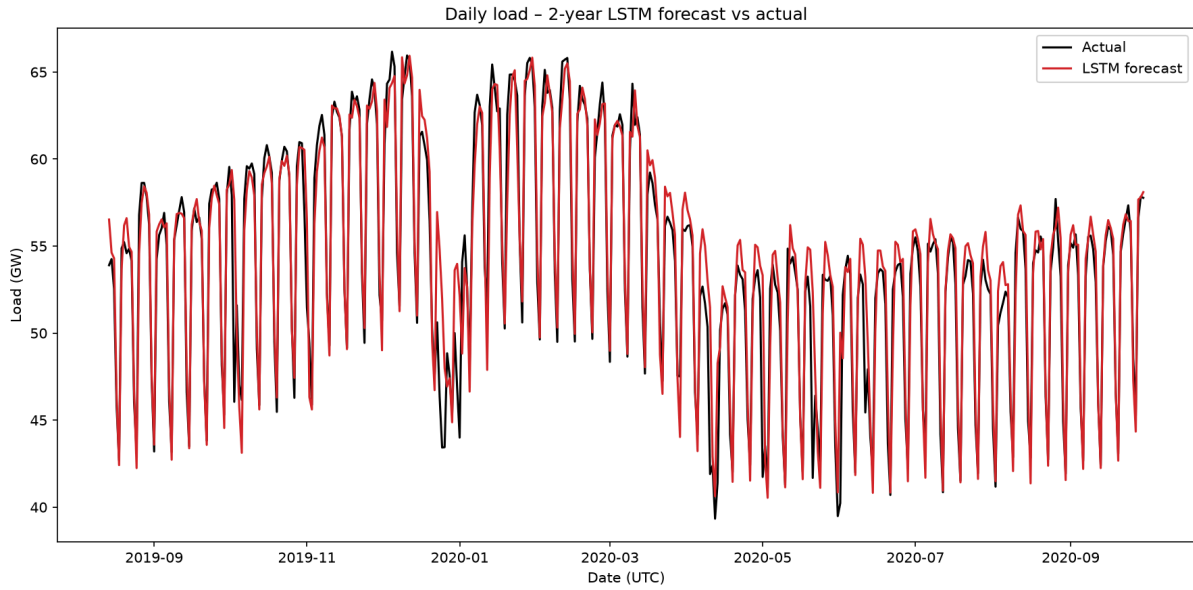


Figure 7. LSTM daily forecast versus actual daily load over the 2-year test horizon (*lstm_daily_2y.png*).

4. Results and critical analysis

4.1 Benchmark model performance

Table 1 summarises the benchmark results for the mean, naïve, seasonal naïve and drift forecasts on the weekly 2-year test period.

Table 1. Benchmark forecasting results (test period).

Model	MAE	RMSE	Bias
Seasonal naïve	3.01	3.80	1.45
Naïve	4.46	4.46	0.40
Mean	4.40	4.40	0.16
Drift	5.12	5.12	2.90

The seasonal naïve model is the strongest of the simple benchmarks because the exploratory analysis showed a clear annual seasonal pattern. The fact that seasonal naïve achieves the lowest MAE and RMSE among the benchmarks indicates that a large proportion of weekly German electricity demand can be explained simply by repeating the demand from the equivalent week in the previous year. This confirms the importance of including a seasonal reference when assessing more complex models: if a candidate model cannot beat seasonal naïve, it is not exploiting additional structure beyond the stable yearly cycle.

4.2 SARIMA and SARIMAX performance

The selected SARIMA model $(0,1,3)(0,1,1)_{52}(0,1,3)(0,1,1)_{52}$ achieves lower error than the weaker benchmarks but does not outperform seasonal naïve. Its RMSE is around 3.8, with positive bias indicating a tendency to over-forecast. Residual ACF and PACF plots suggest that most autocorrelation is removed, but the remaining errors still reflect variations that the linear structure cannot capture.

Adding temperature in the SARIMAX model yields a small improvement in RMSE over SARIMA, confirming that weather explains some deviations from the average seasonal pattern. However, SARIMAX still does not surpass the seasonal naïve benchmark in this study, which implies that the added linear effect of temperature is not sufficient to compensate for the model's limitations in capturing nonlinear relationships.

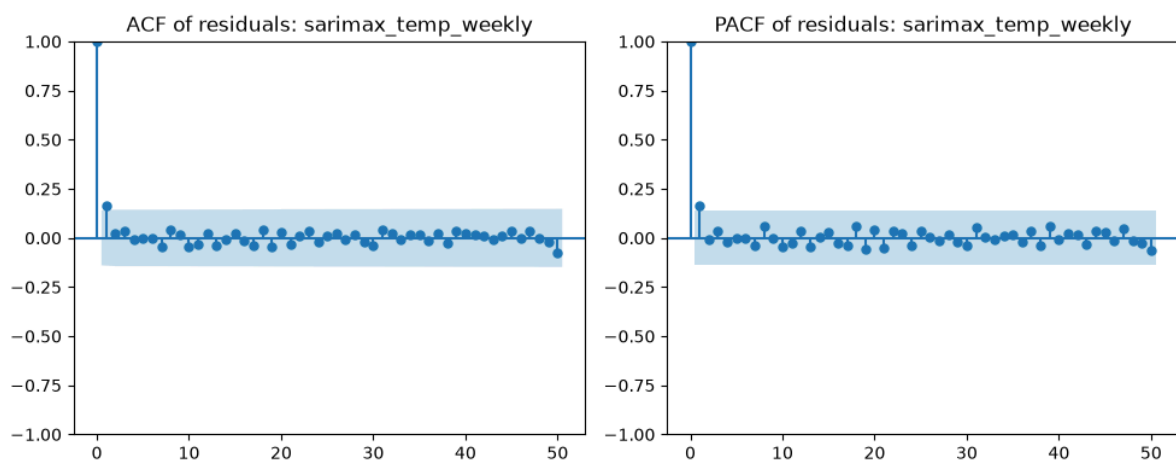


Figure 8. Residual ACF/PACF for SARIMA and SARIMAX models, showing reduced but non-negligible dependence (residual_acf_pacf.png).

4.3 Random Forest performance

The feature-based Random Forest model achieves substantially lower errors than both the benchmarks and the classical time-series models. On the weekly test set, its RMSE is approximately 2.4, representing a reduction of around 20% compared with seasonal naïve, and its bias is smaller, indicating less systematic over- or under-prediction.

Visual inspection shows that the Random Forest forecast tracks peaks and troughs more closely than seasonal naïve, particularly during periods affected by unusual temperature conditions or calendar effects. This supports the interpretation that nonlinear interactions between lagged demand, weather and calendar variables are important for explaining weekly German electricity demand and that the feature-based approach can exploit these interactions effectively.

4.4 LSTM performance

The LSTM model, trained on daily data, delivers the **best overall accuracy** in the study. Its RMSE on the daily test period, when aggregated to the weekly scale, corresponds to an effective weekly RMSE of about 2.1, which is lower than both the Random Forest and all other models. Bias is also relatively small, suggesting that the model captures the level of the series well.

However, the improvement over the Random Forest is modest compared with the gap between Random Forest and seasonal naïve. This indicates that while the LSTM can learn additional nonlinear temporal dependencies, a well-engineered feature-based model already captures most of the predictable structure in this dataset. Moreover, the LSTM’s performance depends on careful hyperparameter tuning and regularisation; without this, the model can overfit, especially given a limited historical window.

4.5 Comparison of all models

Table 2 compares the performance of the main models using the combined metrics file.

Table 2. Comparison of benchmark, SARIMA/SARIMAX, Random Forest and LSTM models (test period).

Model	MAE	RMSE	Bias
LSTM	2.08	2.08	−1.05
RandomForest	2.40	2.40	0.82
Seasonal naïve	3.01	3.01	1.45
SARIMAX+Temp	3.81	3.81	2.29
SARIMA	3.80	3.80	2.34
Mean	4.40	4.40	0.16
Naïve	4.46	4.46	0.40
Drift	5.12	5.12	2.90

The Random Forest and LSTM models provide a **meaningful improvement** over the seasonal naïve benchmark, reducing RMSE by roughly 20–30%. SARIMA and SARIMAX, by contrast, do not beat seasonal naïve on RMSE, which limits their practical benefit despite their interpretability. Importantly, the gains over simpler mean, naïve and drift baselines are much larger but less informative: exceeding a weak baseline is easy and does not justify increased complexity.

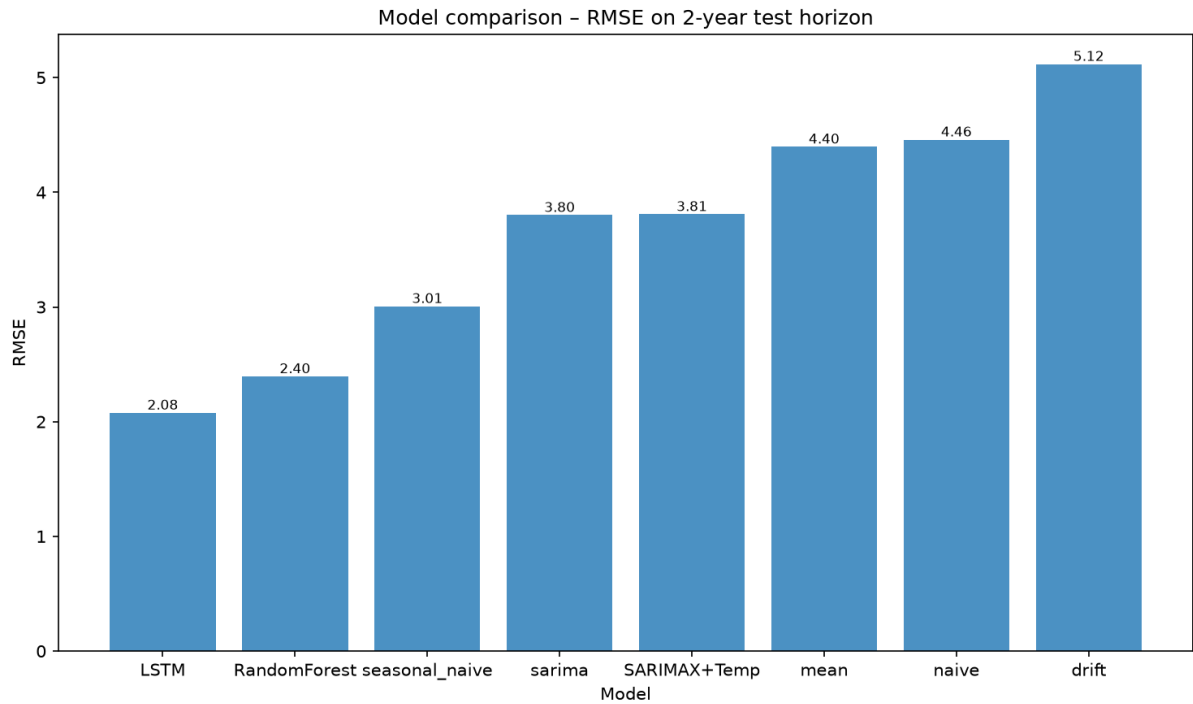


Figure 9. RMSE comparison across all models on the common test horizon (model_comparison_rmse.png).

5. Comparison with realised data

An important test of forecast quality is how well the models perform on realised demand within the test period rather than only in-sample. Visual inspection of the weekly and daily forecast plots shows that the Random Forest and LSTM models capture the overall seasonal pattern and reproduce most major peaks and troughs, with errors concentrated around periods of rapid demand changes.

Comparing these advanced models with the seasonal naïve forecast reveals that the naïve benchmark continues to perform well over large parts of the test period, reflecting the stability of the annual cycle. However, in weeks affected by unusual weather or holiday effects, the Random Forest and LSTM forecasts adapt more quickly, producing smaller errors and demonstrating genuine added value.

In practice, this suggests that for periods where the system behaves “normally”, simple seasonal rules of thumb remain powerful, but when conditions deviate from the historical pattern, models that incorporate covariates and nonlinear structure provide more reliable guidance.

6. Discussion questions

6.1 Comparing models to the seasonal naïve benchmark

The seasonal naïve model is the strongest benchmark because annual seasonality is the dominant characteristic of German electricity demand. Advanced models provide a meaningful improvement

only when they consistently achieve lower MAE and RMSE than the seasonal naïve forecast. In this study, the Random Forest and LSTM satisfy this criterion, while SARIMA and SARIMAX do not, indicating that only the former two exploit additional predictive structure beyond the stable yearly cycle.

6.2 Avoiding data leakage

Temperature lag features were created using only historical information. Variables such as one-week or multi-week lagged temperature were generated by shifting the temperature series backward **before** splitting into training and test sets. Consequently, each prediction used only information available at the forecast origin, and no future temperature values were incorporated into model inputs. This is essential for a fair evaluation because using observed future temperatures directly as predictors would artificially improve performance and invalidate the forecast comparison.

6.3 SARIMAX differencing orders and seasonal period

The differencing orders were selected to achieve approximate stationarity while preserving annual seasonal structure. The seasonal period was set to 52 weeks because the weekly series exhibits a clear yearly cycle, consistent with previous work on electricity load and temperature-sensitive demand. A first-order non-seasonal difference removes the long-term trend, and a first-order seasonal difference at lag 52 removes the annual pattern. ACF/PACF plots and ADF tests on the differenced series support this choice, and the selected SARIMAX model $(0,1,3)(0,1,1)_{52}(0,1,3)(0,1,1)_{52}$ minimises AIC within the considered grid.

6.4 Temperature and calendar covariates

Temperature and calendar variables improve forecast accuracy by explaining deviations from the normal seasonal pattern. In the Random Forest model, these covariates receive substantial importance scores, and the model's superior accuracy indicates that they help explain unusual weeks that SARIMA cannot handle well. In SARIMAX, temperature provides a modest improvement over SARIMA, but not enough to beat seasonal naïve. Calendar effects such as week-of-year and holidays are known in advance, while temperature must be obtained from weather forecasts; therefore these covariates are suitable exogenous predictors in operational forecasting provided reliable upstream forecasts are available. When observed future temperature is used directly in evaluation, the resulting forecasts are better described as explanatory or conditional rather than fully operational.

6.5 Interpretability and complexity

SARIMAX offers the highest interpretability among the advanced models: its parameters map directly to autoregressive, moving-average and seasonal effects, and exogenous coefficients quantify the marginal effect of temperature. Random Forest provides intermediate interpretability; individual trees are complex, but feature importance and partial dependence can still indicate how lagged load and temperature influence forecasts. The LSTM is the least interpretable because its internal states and weights are difficult to explain, even though it produces the lowest errors. In operational settings,

interpretability matters because operators often need to understand **why** a forecast changes, not just the numerical value.

6.6 Recommended operational model

For operational use, the recommended model is the **weekly Random Forest feature-based regressor**. It provides strong forecasting accuracy, achieving a substantial improvement over seasonal naïve, SARIMA and SARIMAX, while remaining simpler to train, retrain and deploy than the LSTM. It also offers some transparency through feature importance analysis, which helps operators understand the impact of temperature and calendar variables. The LSTM achieves the lowest RMSE, but the gain over Random Forest is modest and comes at the cost of higher complexity, more demanding infrastructure and reduced interpretability. In many practical environments, the Random Forest therefore offers the best balance between accuracy, uncertainty awareness (via residual analysis and potential bootstrapping) and long-term maintainability.

7. Limitations and future improvements

Several limitations affect the scope of this study. First, the weekly dataset contains a limited number of observations from 2015–2020; this restricts the information available for training higher-capacity models and makes overfitting more likely. Second, the set of exogenous variables is relatively small: only temperature and basic calendar features are used, whereas electricity demand is also influenced by economic activity, prices and sector-specific factors. Third, the evaluation relies on a single 2-year holdout period; rolling-origin cross-validation or probabilistic scoring rules would provide a more robust assessment of forecast stability over time.

These limitations suggest clear directions for future work. Extending the dataset to include more years or additional countries, modelling hourly load directly with richer calendar structure, and incorporating more detailed weather variables (e.g. heating and cooling degree days) would likely strengthen performance. Hybrid models that combine SARIMAX for linear seasonal structure with a neural network on residuals could also be explored, as well as methods that produce full predictive distributions rather than just point forecasts. Finally, using true weather forecasts instead of realised temperatures would allow a more realistic evaluation of covariate-driven models in an operational context.

8. Conclusion

This case study evaluated benchmark, statistical, feature-based and neural network approaches for forecasting German electricity demand using open-source data and reproducible Python code. Strong annual seasonality made the seasonal naïve benchmark difficult to outperform, highlighting the importance of rigorous baseline comparisons.

The main finding is that both the feature-based Random Forest and the LSTM model deliver meaningful improvements over the seasonal naïve benchmark, with the LSTM achieving the lowest RMSE overall. However, the Random Forest achieves almost comparable accuracy while offering greater interpretability, simpler deployment and lower maintenance costs. Classical SARIMA and

SARIMAX models remain valuable for understanding the series and providing transparent baselines, but they do not match the performance of the nonlinear approaches on this dataset.

For operational use, the Random Forest is recommended as the primary forecasting model, with SARIMAX retained as an interpretable secondary reference and the LSTM as a candidate for environments where maximum accuracy justifies the additional complexity. Overall, the results illustrate that incorporating external variables such as temperature and using flexible modelling approaches can provide valuable improvements in electricity demand forecasting, but that these gains must be weighed against interpretability and maintainability when selecting a model for real-world deployment.

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